
資料視覚化

Data Visualization

Week 7: More Pandas

Split-Apply-Combine

What is Split-Apply-Combine?

- A common data analysis pattern:
 - Split data into groups
 - Apply a function to each group
 - Combine results into a new structure

	species	sepal_length	sepal_width	petal_length	petal_width
0	setosa	5.1	3.5	1.4	0.2
1	setosa	4.9	3.0	1.4	0.2
2	setosa	4.7	3.2	1.3	0.2
3	setosa	4.6	3.1	1.5	0.2
4	setosa	5.0	3.6	1.4	0.2
50	versicolor	7.0	3.2	4.7	1.4
51	versicolor	6.4	3.2	4.5	1.5
52	versicolor	6.9	3.1	4.9	1.5
53	versicolor	5.5	2.3	4.0	1.3
54	versicolor	6.5	2.8	4.6	1.5
100	virginica	6.3	3.3	6.0	2.5
101	virginica	5.8	2.7	5.1	1.9
102	virginica	7.1	3.0	5.9	2.1
103	virginica	6.3	2.9	5.6	1.8
104	virginica	6.5	3.0	5.8	2.2

	species	sepal_length	sepal_width	petal_length	petal_width
	setosa	24.3	16.4	7.0	1.0
	versicolor	32.3	14.6	22.7	7.2
	virginica	32.0	14.9	28.4	10.5

The diagram illustrates the aggregation of data from a large table into a smaller summary table using the SUM function. The large table on the left contains 15 rows of data, grouped by species (setosa, versicolor, virginica). The small table on the right shows the summary statistics (SUM) for each species. Red boxes highlight the columns being summed (sepal_length, sepal_width, petal_length) and the resulting summary values. Red arrows and the word 'SUM' indicate the aggregation process.

Source: <https://medium.com/data-science/how-to-use-the-split-apply-combine-strategy-in-pandas-groupby-29e0eb44b62e>

Joins

ID	X1
1	a1
2	a2

ID	X2
2	b1
3	b2

Inner Join

ID	X1	X2
2	a2	b1

Outer Join

ID	X1	X2
1	a1	NA
2	a2	b1
3	NA	b2

Left Join

ID	X1	X2
1	a1	NA
2	a2	b1

Right Join

ID	X1	X2
2	a2	b1
3	NA	b2

`left_join(x, y)`

1	x1	1	y1
2	x2	2	y2
3	x3	4	y4

`inner_join(x, y)`

1	x1	1	y1
2	x2	2	y2
3	x3	4	y4

A blurry, low-resolution image of a monkey sitting at a desk, typing on a laptop. The monkey is wearing a light-colored shirt. The background is a reddish-brown wall. The text "Time for Python" is overlaid in the center in a large, white, sans-serif font.

Time for Python